

Working title: Parameter efficient fine-tuning ensembles in IMDB- reviews

Problem description

This project is about fine tuning pre-existing Large-Language models such as RoBERTa or DeBERTa utilizing LoRA or full fine-tuning approaches. It will be utilized to automatically categorize IMDB movie reviews into positive and negative reviews to create a snapshot of the sentiment or mood of the audience. As the movie industry is a multi-billion dollar business, it is very important to know how a certain movie is received by the audience, considering the revenue, production of potential sequels or merchandise articles. A brief explanation of the problem being addressed and its importance.

Proposed approach

In this project, a two-level hybrid architecture is proposed to optimize the high computational costs of Large Language Models (LLMs) and maximize sentiment analysis performance. The proposed method integrates Parameter-Efficient Fine-Tuning (PEFT) and Stacking Ensemble techniques across four main phases:

- **Phase 1: Parameter-Efficient Adaptation of Base Models:** Pre-trained RoBERTa and DeBERTa models, which possess high contextual understanding capabilities, are selected to serve as the Level 0 base of the architecture. To minimize hardware and memory usage, the original billions of parameters of these models are frozen. To facilitate the learning process, low-rank LoRA (Low-Rank Adaptation) matrices are integrated into the Transformer-based Attention Blocks of the models. By training only these adapters, less than 1% of the model parameters are updated, ensuring high computational efficiency.
- **Phase 2: Out-of-Fold (OOF) Prediction Generation:** To prevent data leakage and over-fitting problems commonly encountered in ensemble learning, a K-Fold Cross Validation strategy will be utilized. The training data will be divided into 5 equal folds; the base models (RoBERTa and DeBERTa) will be trained on 4 folds and generate probability predictions on the 1 unseen fold. This process will be repeated across the entire dataset to create an unbiased pool of predictions (OOF predictions) for training the Level 1 model.
- **Phase 3: Meta-Learner Training via Stacking:** To capture the synergy between the RoBERTa (robust baseline features) and DeBERTa (disentangled attention) models, a Meta-Learner is employed. The OOF sentiment probabilities obtained in Phase 2 will be horizontally concatenated to form a new feature vector. These vectors will be fed as input into a simpler classifier, such as Logistic Regression or XGBoost, to train the final decision mechanism.
- **Phase 4: Inference Pipeline:** During the live testing phase, a novel IMDB review will sequentially pass through the LoRA-integrated RoBERTa and DeBERTa models to obtain independent probability scores. These scores will then be weighted by the Meta-Learner to perform the final sentiment classification (Positive/Negative) of the text.

Baseline Methods

To comprehensively evaluate the computational efficiency and classification performance of the proposed PEFT and Stacking-based hybrid architecture, the system will be compared against the following three baseline algorithms:

- **Baseline 1: TF-IDF with Support Vector Machine (SVM):** To represent traditional machine learning approaches, Term Frequency-Inverse Document Frequency (TF-IDF) vectorization combined with a Support Vector Machine (SVM) will be utilized. This method serves

as a classical lower-bound reference point, demonstrating the fundamental leap in performance provided by contextual language models compared to sparse statistical representations.

- **Baseline 2: Full Fine-Tuned RoBERTa (Single Model Upper Bound):** To evaluate the computational savings achieved by our proposed LoRA integration, a standard RoBERTa model will be fully fine-tuned (updating all parameters) on the IMDB dataset. This baseline represents the traditional deep learning approach and acts as the upper bound for computational cost. The proposed architecture will be compared against this baseline in terms of training time, peak VRAM usage, and the ratio of trainable parameters.
- **Baseline 3: Single-Model PEFT (RoBERTa + LoRA):** To isolate and evaluate the specific contribution of the Stacking Ensemble phase, a single RoBERTa model fine-tuned solely with LoRA adapters will be used as a third baseline (Ablation Study). By comparing our final Meta-Learner's results against this single-model PEFT baseline, we can quantitatively measure the synergistic performance gain (such as F1-Score and ROC-AUC improvements) achieved by combining diverse model architectures.

Dataset

The dataset used in this project is called "IMDB Dataset of 50k Movie Reviews" and can be downloaded from the following website <https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews> (66MB). It contains 49582 unique instances and only one feature represented by the entire review written by a user. The output is of categorical kind and represents the sentiment expected from the review, so the only two possible categories are positive and negative.

Description of the dataset(s) to be used, including source and size if available.

Evaluation strategy

A first metric to classify the quality of our fine tuned model is the accuracy, measuring the correct predictions out of all predictions. As this metric needs to be handled carefully, other metrics such as Precision, Recall and F1-Score should be evaluated as well to put the accuracy in the right context. Metrics and evaluation methodology (e.g., accuracy, F1-score, RMSE, etc.).

Expected challenges

While the proposed architecture offers significant advantages, several technical challenges are anticipated during the implementation phase. Acknowledging and addressing these challenges is crucial for the robust execution of the project:

- **Handling Long-Text Token Limits:** Transformer-based architectures, such as RoBERTa and DeBERTa, typically possess a strict maximum input sequence limit of 512 tokens. Given that IMDB reviews can frequently exceed this threshold, simple truncation strategies (e.g., arbitrarily cutting only the beginning or the end of the text) may lead to the loss of critical context and sentiment shifts (e.g., plot twists), thereby negatively impacting classification accuracy. Careful evaluation of advanced truncation methods, such as the "Head+Tail" approach, will be required.
- **Data Leakage and Overfitting Risk in Stacking:** A critical vulnerability in the proposed two-level stacking architecture is the potential for data leakage. If the Level 1 Meta-Learner is trained and validated using probability predictions derived from the exact same data used to train the Level 0 models, it will lead to severe overfitting. Consequently, the ensemble model would fail to generalize and perform poorly on unseen test data. Strict adherence to Out-of-Fold (OOF) prediction generation is mandatory to mitigate this structural risk.

- **LoRA Hyperparameter Optimization Overhead:** Achieving the optimal balance between performance and parameter efficiency relies heavily on tuning LoRA-specific hyperparameters, particularly the rank dimension (r), the scaling factor (α), and dropout rates. Since this tuning is inherently an empirical, trial-and-error process, it presents a risk of unforeseen computational resource and time consumption.
- **GPU Memory Management (OOM Constraints):** Although Parameter-Efficient Fine-Tuning (PEFT) drastically reduces the number of *trainable* parameters, the underlying base models (especially DeBERTa) remain architecturally heavy and demand substantial VRAM for forward passes and activations. Loading these models during the ensemble training and inference phases poses a persistent risk of Out-of-Memory (OOM) errors. Meticulous hardware monitoring, careful optimization of training batch sizes, and the potential implementation of gradient accumulation techniques will be essential.